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Low-cost and high-power $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ as a novel cathode with outstanding cyclability for K-ion batteries†

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De/intercalation of K^+ ions results in more severe structural change and volume expansion/shrinkage than that of other monovalent ions such as Li^+ and Na^+ because of the larger K^+ -ion size. Thus, it is important to develop novel cathode materials with stable three-dimensional crystal structure and large K^+ diffusion pathways to prevent severe structural change during repeated K^+ de/intercalation. Here, we introduce $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ composed of three-dimensionally interconnected $[Mn, Fe]O_6$ octahedra and PO_4 tetrahedra as a promising cathode material for K-ion batteries. We demonstrate the outstanding electrochemical properties and reaction mechanism of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ in a K-ion battery system through combined studies using various experimental techniques and first-principles calculation. $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ delivers $\sim 110\text{ mA h g}^{-1}$ with a high average operation voltage of 3.5 V (vs. K^+/K) at C/20 ($1C = 117\text{ mA g}^{-1}$). Even at 5C, its specific capacity is $\sim 85\text{ mA h g}^{-1}$, corresponding to $\sim 77\%$ of the capacity measured at C/20. In addition, $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ exhibits an outstanding capacity retention of $\sim 83\%$ after 300 cycles at C/3 with a high coulombic efficiency of more than 99%. These findings confirm the importance of a stable three-dimensional crystal structure for high-performance K-ion batteries.

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1. Introduction

The development of environmentally friendly and sustainable energy resources is key to preventing the air pollution resulting from the excessive use of fossil fuels; in addition, the advancement of electric vehicles (EVs) and related research is currently a major global issue.^{1–3} Lithium-ion batteries (LIBs) have received great attention as promising energy storage systems (ESSs) for EVs and as power sources for small electronic devices because of their high energy density and power-capability.⁴ However, the limited Li resources in the earth and explosively growing global LIBs demands may result in highly increased production costs for LIBs, which may prevent further application of LIBs for grid-scale ESSs.⁵ Moreover, the various commercialized cathodes for LIBs are mainly

composed of expensive elements, including not only Li but also Co and Ni. Thus, numerous researchers are focused on developing new energy storage systems with lower production costs than LIBs.⁶

Recently, potassium-ion batteries (KIBs) have started to attract great attention as promising alternatives to LIBs for large-scale applications because of the earth abundance of K resources and their alkali-ion-based reaction mechanism, similar to that of LIBs.^{7,8} Moreover, KIBs can have larger theoretical energy densities than other alternatives to LIBs such as Na- and Mg-ion batteries owing to the lower redox potential of K^+/K (-2.93 V vs. the standard hydrogen electrode (SHE)) relative to those of Na^+/Na (-2.71 V vs. SHE) and Mg^{2+}/Mg (-2.27 V vs. SHE).^{9,10} It has also been demonstrated that K^+ ions can be reversibly de/intercalated at the graphite-based electrode materials used as commercial LIBs anodes, indicating that the chemistry of KIBs is similar to that of LIBs.¹¹ However, because of the larger ionic size of K^+ ($\sim 1.38\text{ \AA}$) relative to that of Li^+ ($\sim 0.76\text{ \AA}$), K^+ diffusion is more sluggish than Li^+ diffusion in the same host structure, and K^+ de/intercalation is accompanied by larger structural change than Li^+ de/intercalation.^{12,13} Thus, there is a need for the development of novel cathode materials with not only large ionic diffusion pathways for facile K^+ diffusion but also stable frameworks to suppress the severe volume change during

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charge/discharge. The discovery of appropriate cathode materials is an important issue for the realization of KIBs as competitive alternatives to LIBs in the near future. Various cathode materials for KIBs have been reported and studied, including layered-type compounds, Prussian blue compounds, and polyanion compounds.^{14–21} Layered-type materials have several great merits as promising KIBs cathodes, including large gravimetric capacities and large K^+ diffusion pathways.²² However, it was reported that most of layered-type cathode materials for KIBs exhibit the limited specific capacity at the available voltage range and their poor cycle performances due to large ionic size of K^+ .²³ Thus, we have focused on the development of novel polyanion-based cathode materials with not only large K^+ diffusion pathways but also stable three-dimensional frameworks.^{24,25} Herein, we investigated the various electrochemical and structural properties and the reaction mechanism of $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a new cathode material for KIBs using a combination of first-principles calculation and experiments. The use of not only K elements but also earth-abundant Mn and Fe elements implies potentially high cost competitiveness for the production of this material. Through detailed structural analyses, we show that $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ consists of three-dimensionally interconnected $[\text{Mn}, \text{Fe}]\text{O}_6$ octahedra and PO_4 tetrahedra, indicating its high structural stability. The presence of PO_4 tetrahedra in the structure indicates the higher theoretical redox potential of $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ relative to that of other layered-type cathode materials for KIBs,

resulting from the inductive effect derived from the high electronegativity of phosphorus ions and the weakened transition metal–oxygen (TM–O) bonding. These structural merits contribute to the outstanding electrochemical properties of $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a promising cathode for KIBs. $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ delivered a specific capacity of $\sim 110 \text{ mA h g}^{-1}$ with the high average operation voltage of $\sim 3.5 \text{ V}$ (vs. K^+/K) during charge/discharge at C/20 ($1\text{C} = 117 \text{ mA g}^{-1}$). Even at 5C, up to $\sim 77\%$ of the specific capacity at C/20 was maintained, which is well matched with the low theoretical activation barrier energy for K^+ diffusion. In particular, $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ delivered a capacity retention of $\sim 83\%$ compared with the initial capacity with a high coulombic efficiency of over 99% for 300 cycles at C/3, which is attributed to its small volume change ($\sim 4.35\%$) during charge/discharge, as confirmed by *operando* X-ray diffraction (XRD) analyses.

2. Experimental

2.1 Preparation of $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$

$\text{Na}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ powder was synthesized *via* a solid-state reaction. $\text{Na}_4\text{P}_2\text{O}_7$ (Duksan, 98.5%), $\text{MnC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (Alfa Aesar, 93%), $\text{FeC}_2\text{O}_4 \cdot 2\text{H}_2\text{O}$ (99%, Sigma Aldrich), and $\text{NH}_4\text{H}_2\text{PO}_4$ (Daejung, 98%) in a molar ratio of 1 : 2 : 1 : 2 were used as precursors, and 5 wt% pyromellitic acid (PA, $\text{C}_6\text{H}_{10}\text{O}_2$) (Alfa Aesar, 96%) was additionally prepared. The precursors were ground using high-energy ball milling at 400 rpm for

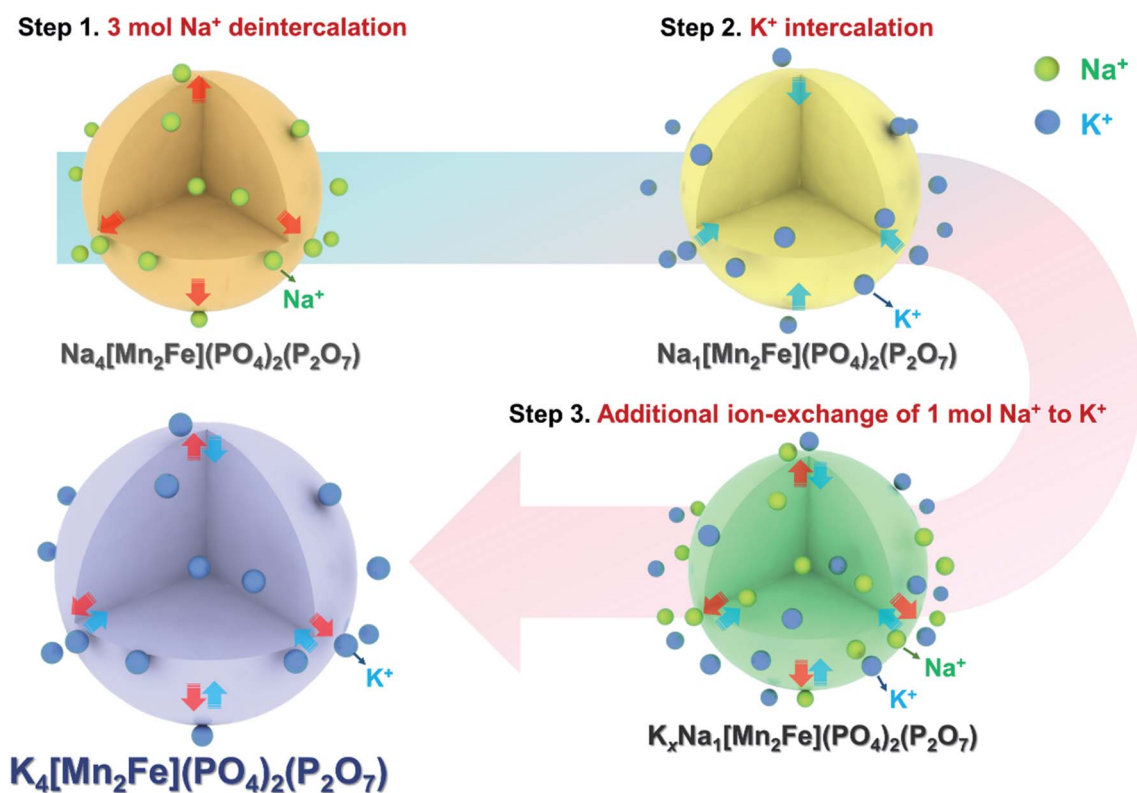


Fig. 1 Schematic illustration of ion-exchange process from $\text{Na}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ to $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$.

12 h. The mixture was calcined at 300 °C for 6 h under Ar flow to remove the organic residues. After calcination, the powder was well ground and high-energy-ball-milled with the 5 wt% PA. The ball-milling jar was filled with Ar to prevent water contamination. The resulting powder was pelletized under a pressure of 200 kg cm⁻² and sintered at 550 °C for 12 h under Ar flow. The sample included approximately 7.7 wt% carbon, as confirmed by thermogravimetric analysis (TGA) (Fig. S1, (ESI[†])). In the ion-exchange process, Na₄[Mn₂Fe](PO₄)₂(P₂O₇) was assembled in a Na-ion cell, and Na ions in the Na₄[Mn₂-Fe](PO₄)₂(P₂O₇) structure were deintercalated by charging to

4.5 V. After the Na⁺ deintercalation, the electrode was reassembled in a K-ion cell, and K ions were intercalated into the crystal structure by discharging to 2.0 V. Then, the K-ion cell was cycled 50 times in the voltage range of 2.0–4.3 V to remove residual Na ions.

2.2 Materials characterization

The K₄[Mn₂Fe](PO₄)₂(P₂O₇) was analyzed using XRD (PANalytical, Empyrean) with Cu Kα radiation (λ = 1.54178 Å) in the 2θ range of 8–80° with a step size of 0.13°. Rietveld refinement of the XRD data was performed using FullProf software. TGA was performed under an argon atmosphere at a heating rate of 10 °C min⁻¹ using a thermogravimetric analyzer (Discovery TGA, USA) to 550 °C. The K₄[Mn₂-Fe](PO₄)₂(P₂O₇) sample was analyzed using the inductively coupled plasma (ICP)-atomic emission spectrometer installed at the National Center for Inter-University Research Facilities (NCIRF) at Seoul National University. The valence states of Mn and Fe in the structure of K₄[Mn₂Fe](PO₄)₂(P₂O₇) were determined from X-ray absorption near edge structure (XANES) analyses performed on beamline 10C at the Pohang Accelerator Laboratory (PAL), South Korea. The Mn K-edge and Fe K-edge spectra were collected in transmission mode with an electron energy of 2.5 GeV and a current of 200 mA, and Mn and Fe reference spectra were simultaneously obtained from Mn and Fe metal foil. Particle size and morphology measurements were performed using scanning electron microscopy (SEM) (Hitachi S-4700) and high-resolution transmission electron microscopy (HR-TEM). Chemical bonding of K and C were analyzed by X-ray photoelectron spectroscopy (XPS, ESCALAB250) at Cooperative Center for Research Facilities (CCRF) at Sungkyunkwan University. *Operando* XRD patterns of K₄[Mn₂Fe](PO₄)₂(P₂O₇) were obtained using an X-ray diffractometer (PANalytical, Empyrean) with Cu Kα radiation (λ = 1.54178 Å) in the 2θ range of 27–35.5° with a step size of 0.026°. An *operando* XRD battery cell

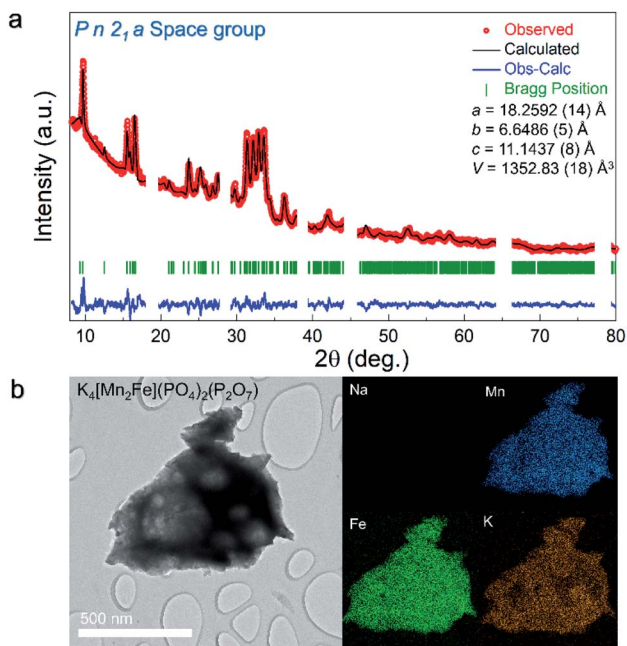


Fig. 2 (a) Refined XRD pattern of K₄[Mn₂Fe](PO₄)₂(P₂O₇). (b) TEM image of K₄[Mn₂Fe](PO₄)₂(P₂O₇) and corresponding EDS mappings of Na, Mn, Fe, and K elements.

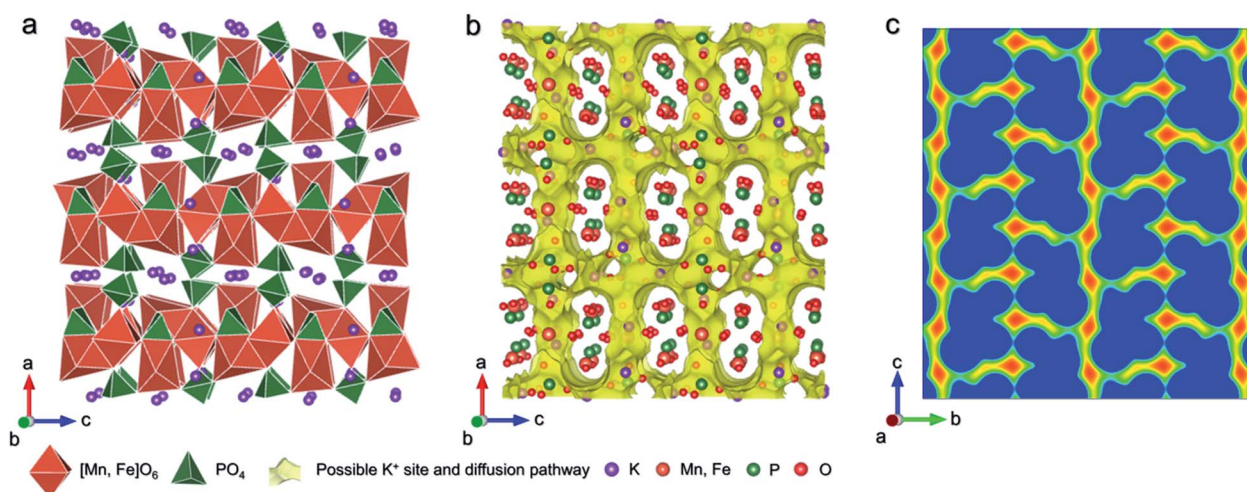


Fig. 3 Crystal structure of K₄[Mn₂Fe](PO₄)₂(P₂O₇). (b) Three-dimensional and (c) two-dimensional (*bc* plane) BVEL analyses of K₄[Mn₂-Fe](PO₄)₂(P₂O₇) with all possible K⁺ diffusion pathways in the crystal structure.

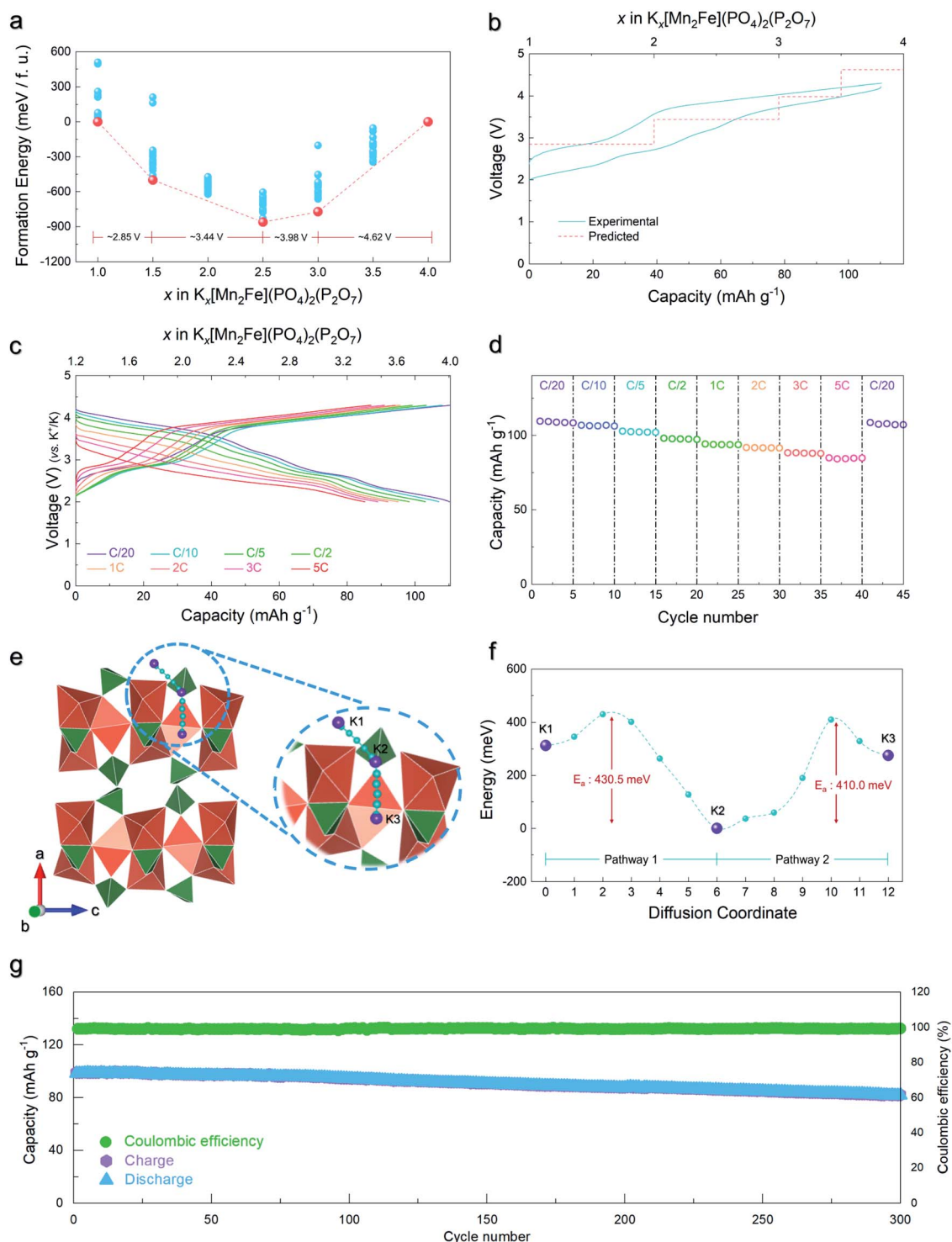


Fig. 4 (a) Convex-hull plot for various $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ configurations ($1 \leq x \leq 4$) with theoretical redox potential. (b) Comparison of predicted redox potential of $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ and its experimentally measured charge/discharge curves for the cycle. (c) Charge/discharge profiles of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ in the range of 2.0–4.3 V at various current rates. (d) Power-capability of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ with several cycles at various current rates. (e and f) Predicted K^+ diffusion motion and predicted activation barrier energy for K^+ diffusion in $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$. (g) Cycle performance of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ at C/3 over 300 cycles in the voltage range of 2.0–4.3 V (vs. K^+/K).

made by PANalytical with PDC TECH was used for the *operando* XRD experiments. To protect the cell system from water, a beryllium (Be) window was used.

2.3 Electrochemical characterization

The $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ electrode was prepared by mixing 75 wt% active material, 15 wt% Super P carbon black, and 10 wt%

polyvinylidene fluoride (PVDF) using *N*-methyl-2-pyrrolidone (NMP) as the solvent. The slurry was applied on Al foil and dried at 100 °C for 24 h under vacuum. After drying, the electrode was punched into a disc (10 ϕ , 3 mg). The mass loading of the electrode was ~ 2 mg cm $^{-2}$. CR2032 cells were prepared using a K $_4$ [Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$) electrode, K metal as the counter electrode, a separator (Whatman GF/F glass fiber), and an electrolyte (0.5 M KPF $_6$ in ethylene carbonate (EC) and diethyl carbonate (DEC) in a volume ratio of 50 : 50). The coin cells were assembled in an Ar-filled glove box. Galvanostatic charge/discharge tests were performed at various C-rates (C/20, C/10, C/5, C/2, 1C, 2C, 3C, and 5C in the range of 2.0–4.3 V) using an automatic battery charge/discharge test system (WBCS 3000, WonATech). In case of galvanostatic intermittent titration technique (GITT) tests, a current density of C/10 was applied for 30 min to measure the closed-circuit voltage (CCV), and it was turned off for 10 min to obtain the quasi-open-circuit voltage (QOCV). The electrochemical impedance spectroscopy (EIS) analyses of the samples were performed through the measurement from 1 MHz to 500 mHz with a 10 mV amplitude using a multichannel impedance analyzer (VSP-300, Bio-Logic, Grenoble, France).

2.4 Computational details

All the density functional theory (DFT) calculations were performed using the Vienna *Ab initio* Simulation Package (VASP).²⁶ We used projector-augmented wave (PAW) pseudopotentials²⁷ with a plane-wave basis set as implemented in VASP. Perdew–Burke–Ernzerhof (PBE) parametrization of the generalized gradient approximation (GGA)²⁸ was used for the exchange–correlation functional. For the DFT calculations, a 4 $\times$ 3 $\times$ 2 k-

point grid was used to calculate theoretical properties of K $_x$ –[Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$). The GGA+U²⁹ method was adopted to address the localization of the d-orbital in Mn and Fe ions, with U_{eff} value of 4.5 eV and 4.3 eV, respectively, determined in previous studies.^{30,31} An appropriate number of k -points and a kinetic energy cutoff of 500 eV were used in all calculations. All the structures were optimized until the force in the unit cell converged to within 0.05 eV Å $^{-1}$.

Nudged elastic band (NEB) calculations³² were performed to determine the activation barrier for K $^+$ diffusion in the K $_4$ –[Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$) structure. To perform the calculations, five intermediate images were generated between each K site. These structures were then calculated using the NEB algorithm³³ with the fixed lattice parameters and free internal atomic positions.

Cluster-assisted statistical mechanics (CASM) software³⁴ was used to generate all the K $^+$ /vacancy configurations for each composition, followed by full DFT calculations on a maximum of 20 configurations with the lowest electrostatic energy for each composition used to obtain the convex hull of K $_4$ [Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$).

3. Results and discussion

3.1 Preparation of K $_4$ [Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$)

The overall procedure for preparing K $_4$ [Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$) is shown in Fig. 1. After deintercalation of ~ 3 mol Na $^+$ from Na $_4$ [Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$) structure during initial charge under the Na-cell system, Na $^+$ /K $^+$ ion-exchange was performed in the K-cell system with K $_x$ Na $_1$ [Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$)|0.5 M KPF $_6$ in EC : DEC (1 : 1 v/v%)|K metal in the voltage range between 2.0

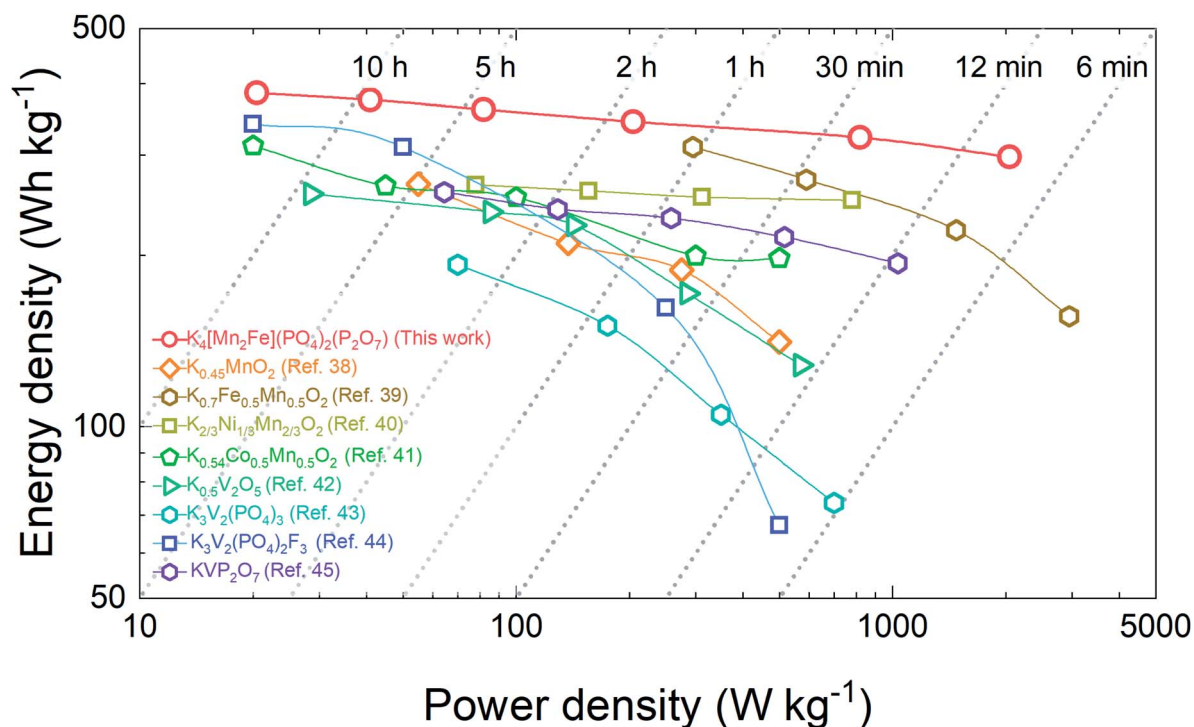


Fig. 5 Comparison of energy and power densities among K $_4$ [Mn $_2$ Fe](PO $_4$) $_2$ (P $_2$ O $_7$) and other reported cathode materials for KIBs.

and 4.3 V (vs. K^+/K) for 50 cycles at 23.4 mA g^{-1} (Fig. S2, (ESI[†])). Repeated charge/discharge under the K-cell system resulted in the ion-exchange from the remaining 1 mol Na^+ to K^+ in the structure. Through these processes, we successfully obtained the $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ phase. In addition, because direct synthesis of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ *via* a solid-state method results in the formation of impurities and several second phases (Fig. S3, (ESI[†])), we applied the ion-exchange process from Na^+ to K^+ in $Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ synthesized *via* a solid-state method for preparation of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$. Moreover, for investigating the component evolution of intermittent phases during ion-exchange process from $Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ to $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$, we performed an inductively coupled plasma-atomic emission spectroscopy (ICP-AES) on the various $K_xNa_{4-x}[Mn_2Fe](PO_4)_2(P_2O_7)$. As shown in Table S1 (ESI[†]), it was verified that the elemental ratio between K and Na in $K_xNa_{4-x}[Mn_2Fe](PO_4)_2(P_2O_7)$ phase is continuously changed during the ion-exchange process without any significant change of Fe and Mn contents in $K_xNa_{4-x}[Mn_2Fe](PO_4)_2(P_2O_7)$ phase. In addition, to investigate the influence of SEI formed during ion-exchange process on the subsequent K-ion storage, we performed the K 2p and C 1s XPS analyses on various $K_xNa_{4-x}[Mn_2Fe](PO_4)_2(P_2O_7)$ ($x = 0, 3, 4$). As shown in Fig. S4 (ESI[†]), it was verified that the XPS spectra of C 1s is retained without any considerable changes during the ion-exchange from Na to K, which implies that SEI is not highly affected during the ion-exchange process of $K_xNa_{4-x}[Mn_2Fe](PO_4)_2(P_2O_7)$ except elemental change from Na to K. Thus, we supposed that high stability of SEI during the ion-exchange process can result in

provided in Table S3 (ESI[†]). The presence of four K^+ atomic sites in the $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ structure was also verified, with Wyckoff positions and atomic coordinates (x, y, z) of 4a (0.500, 0.751, 0.978), 4a (0.289, 0.853, 0.779), 4a (0.394, 0.422, 0.259), and 4a (0.487, 0.714, 0.548), indicating that there are sufficient atomic sites for ~ 3.8 mol K^+ de/intercalation in the $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ structure. Moreover, the TEM and energy-dispersive X-ray spectroscopy (EDS) elemental mapping analyses revealed that Mn, Fe, and K elements were homogeneously dispersed in the $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ particles with an average particle size of ~ 600 nm and that the signal of Na element was negligible (Fig. 2b). The total elemental ratio of Mn : Fe : K based on the TEM-EDS analyses was estimated to be $\sim 2 : 1.003 : 3.805$, which is in good agreement with the inductively coupled plasma-atomic emission spectroscopy (ICP-AES) result (Table S1, (ESI[†])). The average particle size of ~ 600 nm was also confirmed by SEM analyses (Fig. S6, (ESI[†])). Based on the structural information obtained from Rietveld refinement with XRD analyses, the crystal structure of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ is shown in Fig. 3a. The structure consists of $[(Mn, Fe)O_6-PO_4]_\infty$ blocks along the bc plane, with P_2O_7 groups connected with these blocks along the a -axis, resulting in not only a stable three-dimensional framework but also the large ionic diffusion pathways. Moreover, we performed the bond-valence energy landscape (BVEL) analyses using the Bond_str program in the FullProf package and predicted the possible K^+ positions and predicted the possible K^+ positions and diffusion pathways in the structure through three- and two-dimensional BVEL views (Fig. 3b and c).^{35,36}

$$V = - \frac{E[K_{x_2}[Mn_2Fe](PO_4)_2(P_2O_7)] - E[K_{x_1}[Mn_2Fe](PO_4)_2(P_2O_7)] - (x_2 - x_1)E[K]}{(x_2 - x_1)F} \quad (1)$$

easy and smooth K-ion storage at the fully ion-exchanged $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ phase. As shown in Fig. S5 (ESI[†]), it was verified through XRD and TEM analyses that the pure $Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ phase was successfully prepared with an average particle size of ~ 600 nm and that the Mn, Fe, and Na elements were homogeneously distributed in the particles.

3.2 Structural and morphological information of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$

XRD analyses and Rietveld refinement of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ confirmed the orthorhombic crystal structure of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ with a space group $Ph2_1a$ (Fig. 2a). No impurities or second phases were detected in the XRD pattern of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$. In addition, the lattice parameters of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ were observed to increase after ion-exchange from Na^+ to K^+ in the $Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ structure because of the larger ionic size of K^+ (~ 1.37 Å) relative to that of Na^+ (~ 1.02 Å) (Table S2, (ESI[†])). The low reliability factors ($R_p = 2.19\%$, $R_t = 2.88\%$, $R_f = 2.70\%$, and $\chi^2 = 2.58\%$) indicate the high accuracy of the Rietveld refinement results. The detailed structural information of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$, including the atomic positions, B_{iso} values, and occupancies, is

3.3 Electrochemical properties of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$

To predict and understand the theoretical properties of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$, we performed first-principles calculation. The K^+ /vacancy configurations for $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ ($1 \leq x \leq 4$) were generated using CASM software,³⁴ and we predicted the formation energy for each configuration. The formation energies of the $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ ($1 \leq x \leq 4$) configurations were arranged in the form of a convex hull plot (Fig. 4a). The theoretical redox potentials of $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ were calculated using the following equation:

Here, V is the average redox potential in the compositional range, E is the theoretical formation energy of the most stable configuration of each composition, and F is the Faraday constant. It was predicted that ~ 3 mol K^+ ions could be de/intercalated at $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ in the available voltage window for KIBs, and the existence of several intermittent phases between $K_1[Mn_2Fe](PO_4)_2(P_2O_7)$ and $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ implies the single-phase reaction during charge/discharge. As shown in Fig. 4b, it was confirmed that the theoretically predicted electrochemical properties of $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ were well matched with the electrochemically tested charge/discharge profiles. At C/20 ($1C = 117$ mA g^{-1}), $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$

delivers the large specific capacity of $\sim 110 \text{ mA h g}^{-1}$, corresponding to de/intercalation of $\sim 2.8 \text{ mol K}^+$ with an average operation voltage of $\sim 3.5 \text{ V}$ (vs. K^+/K). In addition, Fig. S7 (ESI[†]) presents the cyclic voltammetry (CV) analyses of $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ with a scan rate of at 1 mV s^{-1} at the voltage range of 2.0–4.3 V (vs. K^+/K). Especially, the average operation voltage of $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ is higher than that of $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$ ($\sim 3.32 \text{ V}$ (vs. K^+/K)),³⁷ which indicates that $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ has a merit of high energy density compared to $\text{K}_4\text{Fe}_3(\text{PO}_4)_2(\text{P}_2\text{O}_7)$. Moreover, even at 5C, up to $\sim 77\%$ of the

specific capacity measured at C/20 was maintained, indicating the outstanding power-capability of $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ as a promising cathode for KIBs (Fig. 4c and d). We also predicted the theoretical activation barrier energies for K^+ diffusion for the $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ structure using the NEB method based on first-principles calculation. As shown in Fig. 4e and f, the required activation barrier energies along the K1–K2 and K2–K3 pathways were verified to be ~ 430.5 and $\sim 410.0 \text{ meV}$ respectively, which implies that the sufficiently large space in the $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ structure allows facile K^+ diffusion despite

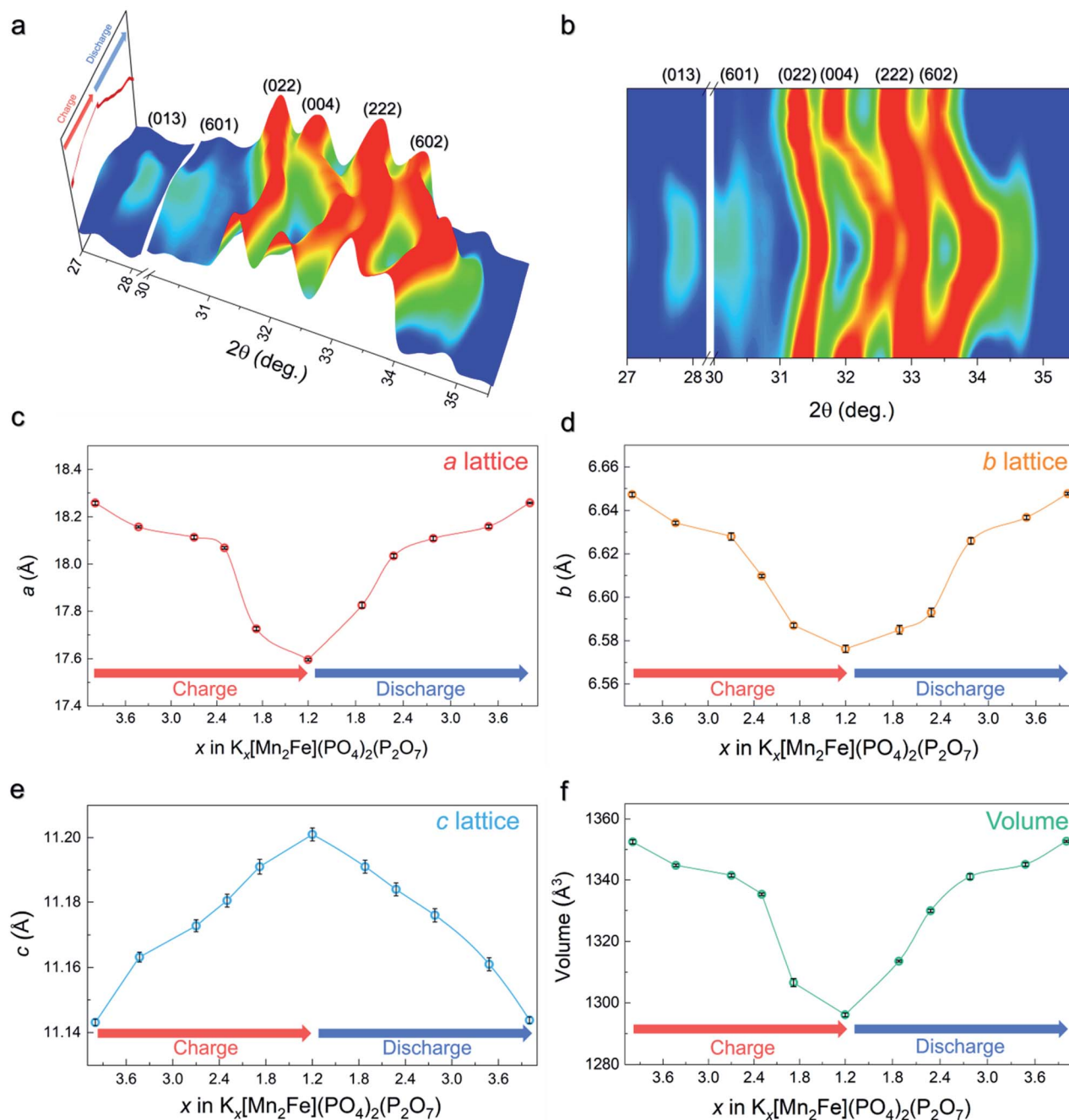


Fig. 6 (a) and (b) Operando XRD patterns of $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ ($1.2 \leq x \leq 4.0$) measured at a current rate of C/10 in the voltage range of 2.0–4.3 V (vs. K^+/K). (c–f) Change in lattice parameters a , b , c and volume of $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ during K^+ de/intercalation in the structure.

the large ionic radius of K^+ . Moreover, $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ exhibited a capacity retention of $\sim 83\%$ compared with the initial capacity after 300 cycles at C/3, with a high coulombic efficiency of over $\sim 99\%$ (Fig. 4g). In case of the cyclability test at 5C, the specific capacity of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ was also well retained up to $\sim 84\%$ of the initial capacity after 200 cycles (Fig. S8, (ESI †)). The excellent cycle performance of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ was also verified in a full-cell test using $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ and hard carbon as the cathode and anode, respectively. Before the full-cell assembly, the hard carbon anode was pre-cycled in the voltage range between 0.01 and 2 V (vs. K^+/K) to eliminate the irreversible initial discharge capacity. Fig. S9 (ESI †) shows that the full-cell exhibited long-term cycle performance with $\sim 82\%$ retention of the initial capacity after 200 cycles at C/5 and a high coulombic efficiency of more than 99%. In addition, we compared the various electrochemical performances among $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ and the previously reported cathode materials for KIBs.^{38–45} Fig. 5 and Table S4 (ESI †) presents that $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ exhibits high energy density, outstanding power-capability and stable cycle-performance compared to other cathode materials. In addition, Fig. S10a (ESI †) presents the electrochemical tests of

$Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ under the KIB system with the various current rates from C/20 to 5C ($1C = 117 \text{ mA g}^{-1}$). Even at 5C, the specific capacity is maintained up to $\sim 70\%$ of the capacity measured at C/20. Moreover, $Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ delivers capacity retention of $\sim 85\%$ with high coulombic efficiency above $\sim 99\%$ after 100 cycles at 1C under the KIB system (Fig. S10b, (ESI †)). These results indicate that $Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ exhibits outstanding electrochemical performances for K-ion storage. Fig. S11 (ESI †) presents the various electrochemical properties of $Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ and $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ under the KIB system, such as cyclic voltammetry (CV), galvanostatic intermittent titration technique (GITT) and electrochemical impedance spectroscopy (EIS). It was verified that there is slight different electrochemical reaction during initial charging/discharging process of $Na_4[Mn_2Fe](PO_4)_2(P_2O_7)$ and $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ under the KIB system, which results from the their difference of initially deintercalated ions and the existence and non-existence of Na^+ in the structure.

3.4 Reaction mechanism of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$

The excellent cyclability of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ in the KIB system was thought to arise from the small structural change by

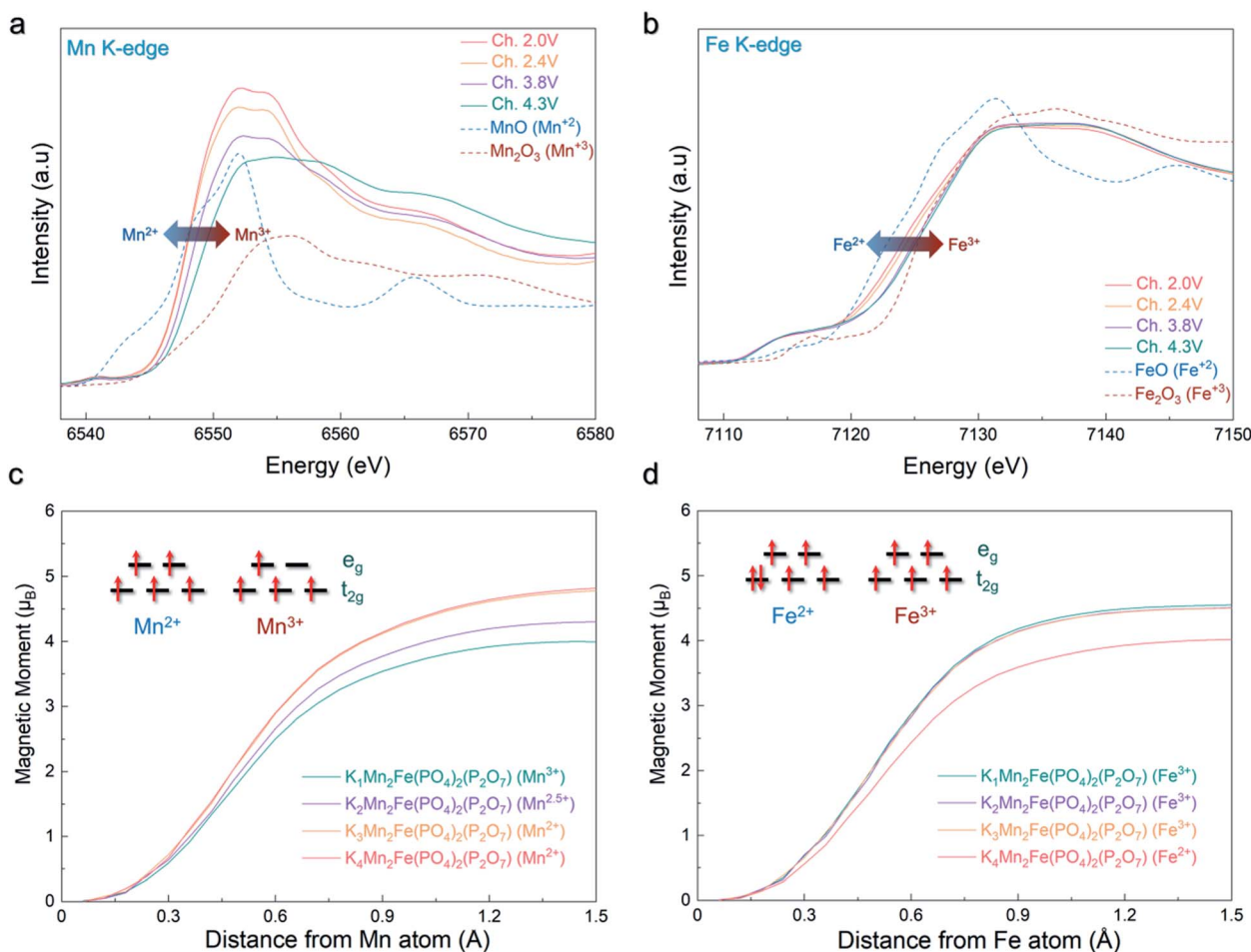


Fig. 7 *Ex situ* (a) Mn and (b) Fe K-edge XANES analyses of $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ ($1.2 \leq x \leq 4.0$). (c and d) Comparison of integrated spin moments on Mn ion and Fe ion between $K_1[Mn_2Fe](PO_4)_2(P_2O_7)$ and $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$.

the three-dimensionally interconnected $[\text{Mn}, \text{Fe}]\text{O}_6$ octahedra and PO_4 tetrahedra during charge/discharge. To investigate the volume change of $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ during K^+ de/intercalation, we performed *operando* XRD analyses (Fig. 6a and b). The full *operando* XRD (*o*-XRD) patterns are also presented in Fig. S12 (ESI†). No phase transition or formation of impurities was detected in the *o*-XRD patterns. The XRD peaks corresponding to (022), (004), (222), (602), *etc.* planes were monotonously shifted toward higher (or lower) 2θ during charge (or discharge), indicating the reversible single-phase reaction of $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ during K^+ de/intercalation. Moreover, it was verified through Rietveld refinement of the *o*-XRD patterns that the lattice parameters of $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ were

continuously increased/decreased during charge/discharge (Fig. 6c–f). In particular, we confirmed that the total volume change during K^+ de/intercalation was only $\sim 4.35\%$ despite the large ionic radius of K^+ . It was speculated that the small volume change and lack of phase transition of $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ during K^+ de/intercalation resulted from the stable three-dimensional framework and large pathways for facile K^+ diffusion. In addition, we investigated the structural morphological change of $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ after numerous charge/discharge cycles through the *ex situ* SEM, TEM, and XRD analyses. Fig. S13 (ESI†) shows the negligible differences in the crystal structure and morphology of the pristine and 300-cycled samples, indicating the stably retained crystal structure of

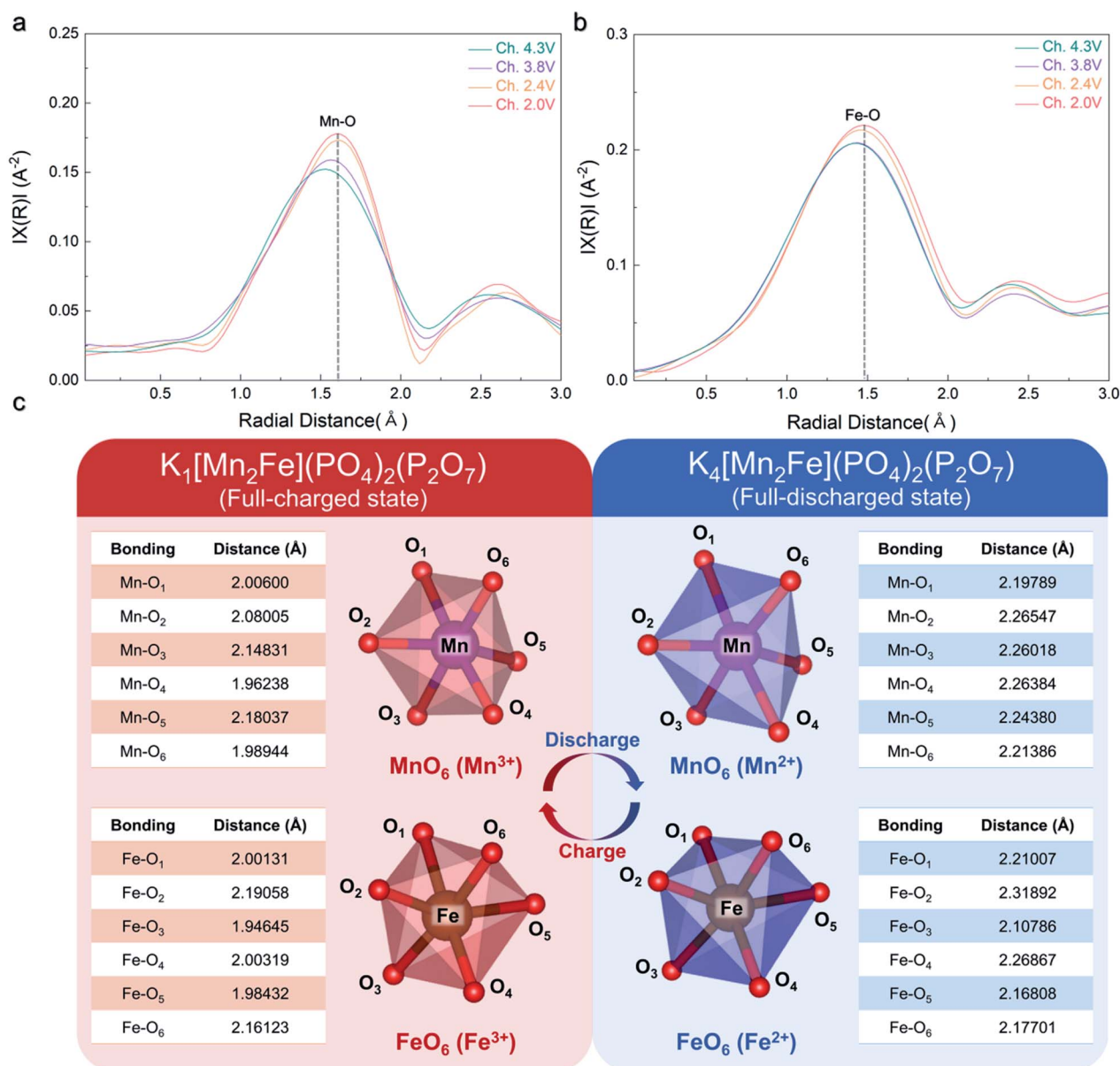


Fig. 8 *Ex situ* EXAFS analyses of $\text{K}_x[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ ($1.2 \leq x \leq 4.0$) with the (a) Mn–O bonding distance and (b) Fe–O bonding distance. (c) Comparison of predicted Mn–O and Fe–O bond distance of MnO_6 and FeO_6 octahedra between $\text{K}_1[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ and $\text{K}_4[\text{Mn}_2\text{Fe}](\text{PO}_4)_2(\text{P}_2\text{O}_7)$ determined through first-principles calculation.

$K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ despite the repeated volume change for 300 cycles.

The change of the oxidation states of the Fe and Mn elements of $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ during charge/discharge was investigated using *ex situ* XANES. As shown in Fig. 7a and b, the Fe K-edge and Mn K-edge spectra were shifted toward higher energy during charge and returned to the original states after charge, indicating the occurrence of Fe^{2+}/Fe^{3+} and Mn^{2+}/Mn^{3+} redox reactions during K^+ de/intercalation at $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$. Moreover, the Fe^{2+}/Fe^{3+} redox reaction mainly occurred when initially charged to 2.0 V (vs. K^+/K) with negligible change of the Mn K-edge. During further charge from 3.1 to ~ 4.3 V (vs. K^+/K), no change in the Fe K-edge was observed, and the Mn K-edge was shifted toward higher energy. The Fe^{2+}/Fe^{3+} and Mn^{2+}/Mn^{3+} redox reactions at $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ were also determined through first-principles calculation. As shown in Fig. 7c and d, we predicted theoretical integrated spin moments of Fe and Mn ions at various $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ compositions. The total spin count of Fe ions changed from 5 to 4 and that of Mn ions remained the same during 1 mol K^+ deintercalation from $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ to $K_3[Mn_2Fe](PO_4)_2(P_2O_7)$. After further K^+ deintercalation from $K_3[Mn_2Fe](PO_4)_2(P_2O_7)$, it was verified to increase the total spin count of Mn ions from 4 to 5 with an unchanged total spin count of Fe ions. These first-principles calculation results indicate that ~ 2.8 mol K^+ can be de/intercalated at $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ via Fe^{2+}/Fe^{3+} and Mn^{2+}/Mn^{3+} redox reactions, which is consistent with the *ex situ* XANES results.

Furthermore, we performed Fourier-transform (FT) analyses of the EXAFS to confirm the local structural changes of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ during charge/discharge. As shown in Fig. 8a and b, the Mn–O and Fe–O bonding distances in the $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ structure slightly and monotonously changed during charge/discharge. In particular, although it has been reported that Mn^{3+} ions experience Jahn–Teller distortion,⁴⁶ no severe changes such as peak splitting were detected in the Mn–O bonding environment during the Mn^{2+}/Mn^{3+} redox reaction. The small local structural change of $K_x[Mn_2Fe](PO_4)_2(P_2O_7)$ during charge/discharge was also confirmed through first-principles calculation. Fig. 8c compares the theoretical Fe–O and Mn–O bonding environments of $K_1[Mn_2Fe](PO_4)_2(P_2O_7)$ and $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$, revealing not only the small variation in the Fe–O and Mn–O bonding environments despite de/intercalation of numerous K^+ ions but also the similar Mn–O bonding distances at the $Mn^{3+}O_6$ octahedra, indicating suppressed Jahn–Teller distortion of Mn^{3+} . These results provided additional reasons for the outstanding cyclability of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ as a promising cathode for KIBs.

4. Conclusions

Through first-principles calculation and the use of the BVLE method based on the structural information, we predicted various theoretical properties of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ such as the possible atomic sites and diffusion pathways for K^+ ions, the redox potentials, the activation barrier energies for K^+ diffusion, and the reaction mechanism occurring during K^+ de/

intercalation. Our findings indicate that $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ can be applied as a promising cathode for KIBs. At C/20 ($1C = 117$ mA g^{-1}), $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ delivered a specific capacity of ~ 110 mA h g^{-1} with an average operation voltage of ~ 3.5 V (vs. K^+/K), corresponding to 3.8 mol K^+ de/intercalation in the structure. In addition, even at 5C, up to $\sim 77\%$ of the specific capacity measured at C/20 was retained. This excellent power-capability of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ is attributed to the low activation barrier energy for facile K^+ diffusion in the structure. Moreover, for 300 cycles at C/3, $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ exhibited a capacity retention of $\sim 83\%$ compared with the initial capacity with a high coulombic efficiency of over 99%, which resulted from the small structural change of $K_4[Mn_2Fe](PO_4)_2(P_2O_7)$ composed of the three-dimensionally interconnected framework during charge/discharge. Our findings provide outstanding insight for the development of low-cost cathode materials with superior electrochemical properties not only for the KIBs system but also for other rechargeable cathode materials.

Conflicts of interest

There are no conflicts of interest to declare.

Acknowledgements

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